Title:

MLOps OPPE-2 — End-to-End ML System

Subtitle:

Heart Disease Prediction

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Problem Statement & Dataset

- Binary classification: Heart Disease (yes / no)
- Dataset from UCI Heart Disease repository
- 14 clinical features (age, cholesterol, ECG, etc.)

Deliverable 1: Repository Setup (COMPLETED)

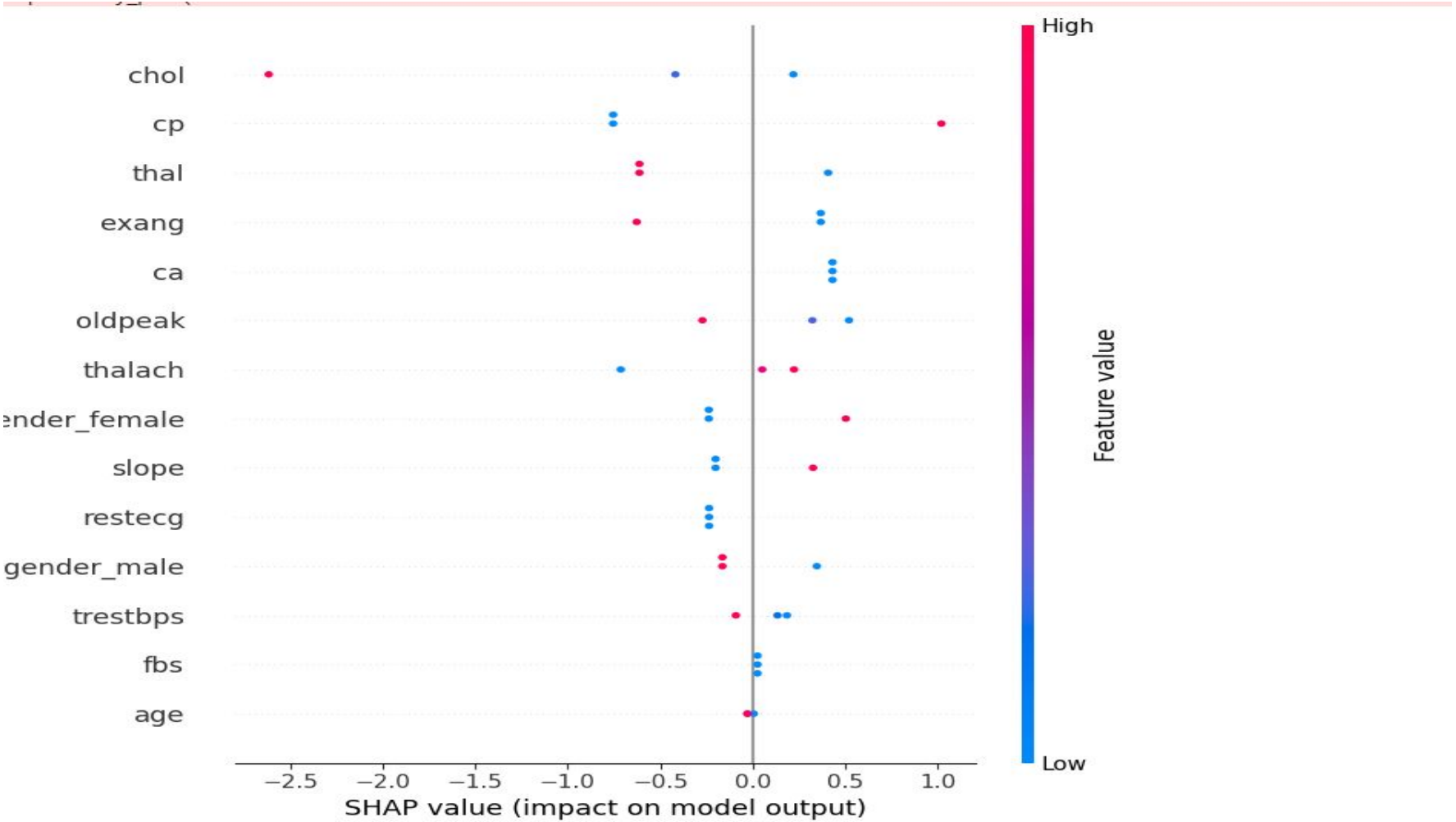
Text:

- Private GitHub repository created
- Naming format followed:
  `23f3001764_IITMBS_MLOPS_OPPE2`
- IITMBSMLOps added as collaborator

Deliverable 2: Explainability (SHAP) ✅

- SHAP used to explain **false negatives**
- Identified features contributing to missed predictions
- Top factors: cholesterol, chest pain, thal, exang, oldpeak

This SHAP plot explains why some heart disease cases were not predicted. Features like cholesterol, chest pain type, and exercise-induced angina had strong influence


chol
cp
thal
exang
ca
oldpeak
thalach
ender_female
slope
restecg
gender_male
trestbps
fbs
age
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
SHAP value (impact on model output)
High
Feature value
Low

Deliverable 3: Fairness Testing (Fairlearn) ✅

- Sensitive attribute: **Age**
- Age bucketed into 20-year bins
- Metrics used:
    - Accuracy by group
    - Demographic Parity Difference
    - Equalized Odds Difference

Fairness was evaluated across age groups using Fairlearn. Minor disparities were observed, indicating acceptable but improvable fairness

STEP 5 — FAIRNESS METRICS USING FAIRLEARN

[55]:
```
metric_frame = MetricFrame(
    metrics=accuracy_score,
    y_true=y_test_bin,
    y_pred=y_pred_bin,
    sensitive_features=age_groups
)

metric_frame.by_group
```

[55]:
```
age
20-39    1.000000
40-59    0.820513
60-79    0.722222
Name: accuracy_score, dtype: float64
```

[56]:
```
from fairlearn.metrics import demographic_parity_difference

dp_diff = demographic_parity_difference(
    y_true=y_test_bin,
    y_pred=y_pred_bin,
    sensitive_features=age_groups
)

dp_diff
```

[56]:
```
0.3055555555555556
```

[57]:
```
from fairlearn.metrics import equalized_odds_difference

eo_diff = equalized_odds_difference(
    y_true=y_test_bin,
    y_pred=y_pred_bin,
    sensitive_features=age_groups
)

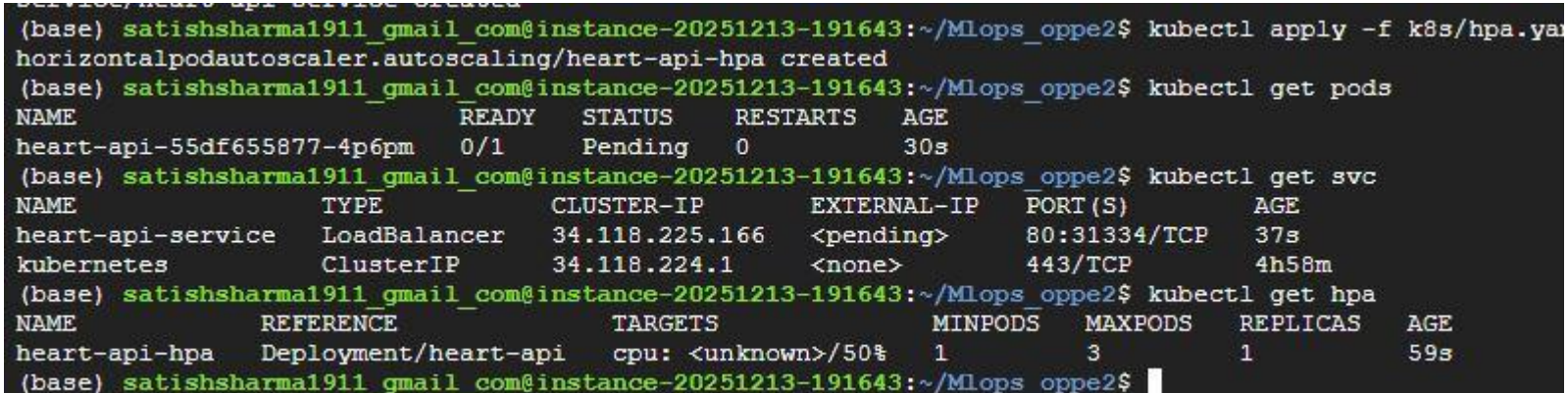
eo_diff
```

[57]:
```
0.375
```

Deliverable 4: API + Docker + Kubernetes (4 MARKS) 🔥

- Model deployed as FastAPI
- Dockerized and pushed to GCP
- Deployed on GKE
- LoadBalancer used for external access
- HPA enabled (max pods = 3)

The model is deployed on GKE with autoscaling enabled. The API is externally accessible via LoadBalancer and scales up to three pods.



```
    "staging-k8s.gcr.io": "gcloud",
    "marketplace.gcr.io": "gcloud"
  }
}
Adding credentials for: us-central1-docker.pkg.dev
After update, the following will be written to your Docker config file located at
[/home/satishsharma1911_gmail_com/.docker/config.json]:
 {
  "credHelpers": {
    "gcr.io": "gcloud",
    "us.gcr.io": "gcloud",
    "eu.gcr.io": "gcloud",
    "asia.gcr.io": "gcloud",
    "staging-k8s.gcr.io": "gcloud",
    "marketplace.gcr.io": "gcloud",
    "us-central1-docker.pkg.dev": "gcloud"
  }
}

Do you want to continue (Y/n)?  Y

Docker configuration file updated.
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ gcloud auth configure-docker us-central1-docker.pkg.dev
WARNING: Your config file at [/home/satishsharma1911_gmail_com/.docker/config.json] contains these credential helper entries:

{
  "credHelpers": {
    "gcr.io": "gcloud",
    "us.gcr.io": "gcloud",
    "eu.gcr.io": "gcloud",
    "asia.gcr.io": "gcloud",
    "staging-k8s.gcr.io": "gcloud",
    "marketplace.gcr.io": "gcloud",
    "us-central1-docker.pkg.dev": "gcloud"
  }
}
Adding credentials for: us-central1-docker.pkg.dev
gcloud credential helpers already registered correctly.
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ docker tag heart-api us-central1-docker.pkg.dev/<PROJECT_ID>/mlops-repo/heart-api:latest
-bash: PROJECT_ID: No such file or directory
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ docker tag heart-api us-central1-docker.pkg.dev/carbon-zone-473306-g5/mlops-repo/heart-api:latest
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ docker push us-central1-docker.pkg.dev/^Clops-repo/heart-api:latest
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ docker push us-central1-docker.pkg.de
v/carbon-zone-473306-g5/mlops-repo/heart-api:latest
The push refers to repository [us-central1-docker.pkg.dev/carbon-zone-473306-g5/mlops-repo/heart-api]
16288cf9e3df: Pushed
83adab69d899: Pushed
841ca3afb7b7: Pushed
503d428e389a: Pushed
0f4f6b519aaa: Pushed
17bbb46fa426: Pushed
dc27ff3b7431: Pushed
557139ca12a0: Pushed
77a2b55fbe8b: Pushed
latest: digest: sha256:f8f66b1298c78666cd6ef3b5ad106cb99fc6f916cb43486a987cacc62b5549ff size: 2201
```

```
C:\Users\Sakshi>curl -X POST http://136.119.155.126/predict ^
More?   -H "Content-Type: application/json" ^
More?   -d "{\"age\":63,\"gender\":\"male\",\"cp\":3,\"trestbps\":145,\"chol\":233,\"fbs\":1,\"restecg\":0,\"thalach\":1
50,\"exang\":0,\"oldpeak\":2.3,\"slope\":0,\"ca\":0,\"thal\":1}"
{"prediction":"yes","probability":0.8725883252020576}
C:\Users\Sakshi>
C:\Users\Sakshi>curl -X POST http://136.119.155.126/predict ^
More?   -H "Content-Type: application/json" ^
More?   -d "{\"age\":35,\"gender\":\"female\",\"cp\":0,\"trestbps\":120,\"chol\":180,\"fbs\":0,\"restecg\":1,\"thalach\"
:170,\"exang\":0,\"oldpeak\":0.2,\"slope\":1,\"ca\":0,\"thal\":2}"
{"prediction":"yes","probability":0.9470722464929916}
C:\Users\Sakshi>curl http://136.119.155.126/
{"status":"ok"}
```

Deliverable 5: Per-Sample Prediction & Logging (2 MARKS)

- 100 random samples generated
- Each sent as individual API request
- Logging enabled:
    - input
    - prediction
    - probability
    - latency
- Observed via Kubernetes logs

Each prediction request is logged with latency and prediction output, demonstrating observability at inference time.

```
Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-packages (2.32.5)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/conda/lib/python3.10/site-packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/conda/lib/python3.10/site-packages (from requests) (2.6.1)
Requirement already satisfied: certifi>=2017.4.17 in /opt/conda/lib/python3.10/site-packages (from requests) (2025.11.12)
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ python generate_and_predict.py
1: {'prediction': 'no', 'probability': 0.3534521332621423, 'latency_sec': 0.24967527389526367}
2: {'prediction': 'no', 'probability': 0.3422574331715365, 'latency_sec': 0.227014303207397746}
3: {'prediction': 'no', 'probability': 0.4387797059534745, 'latency_sec': 0.015807151794433594}
4: {'prediction': 'no', 'probability': 0.11936287680010378, 'latency_sec': 0.034892797470092777}
5: {'prediction': 'no', 'probability': 0.021107550164611508, 'latency_sec': 0.024760007858276367}
6: {'prediction': 'yes', 'probability': 0.8537913304151751, 'latency_sec': 0.022184133529663086}
7: {'prediction': 'yes', 'probability': 0.9008404572578759, 'latency_sec': 0.20246052742004395}
8: {'prediction': 'yes', 'probability': 0.9134108070588103, 'latency_sec': 0.016958236694335938}
9: {'prediction': 'no', 'probability': 0.08777578903591914, 'latency_sec': 0.019358396530151367}
10: {'prediction': 'yes', 'probability': 0.9614148197104451, 'latency_sec': 0.020756483078002937}
11: {'prediction': 'yes', 'probability': 0.5357044986659967, 'latency_sec': 0.014662742614746094}
12: {'prediction': 'yes', 'probability': 0.9614963883156135, 'latency_sec': 0.018270969390869144}
13: {'prediction': 'yes', 'probability': 0.7831960364461505, 'latency_sec': 0.015782594680786133}
14: {'prediction': 'yes', 'probability': 0.819001112190821, 'latency_sec': 0.018836975097656257}
15: {'prediction': 'no', 'probability': 0.3859014681389736, 'latency_sec': 0.019167900085449227}
16: {'prediction': 'no', 'probability': 0.4701397470339982, 'latency_sec': 0.056708574295043945}
17: {'prediction': 'yes', 'probability': 0.619279348103768, 'latency_sec': 0.017599344253540044}
18: {'prediction': 'yes', 'probability': 0.9849471535548056, 'latency_sec': 0.015350580215454102}
19: {'prediction': 'no', 'probability': 0.28161499750373975, 'latency_sec': 0.058118581771850586}
20: {'prediction': 'yes', 'probability': 0.9949077098535398, 'latency_sec': 0.041539907455444336}
21: {'prediction': 'no', 'probability': 0.382312284558457675, 'latency_sec': 0.028100013732910156}
22: {'prediction': 'yes', 'probability': 0.7365963602241395, 'latency_sec': 0.06450915336608887}
23: {'prediction': 'yes', 'probability': 0.8727994412303813, 'latency_sec': 0.028645753860473633}
24: {'prediction': 'yes', 'probability': 0.9820335021524098, 'latency_sec': 0.038182258605957031}
25: {'prediction': 'yes', 'probability': 0.632996651595783, 'latency_sec': 0.05236005783081055}
26: {'prediction': 'yes', 'probability': 0.7378528590548885, 'latency_sec': 0.059209585189819336}
27: {'prediction': 'yes', 'probability': 0.7877746371749639, 'latency_sec': 0.042913913726806641}
28: {'prediction': 'no', 'probability': 0.14922420253629828, 'latency_sec': 0.07382297515869141}
29: {'prediction': 'yes', 'probability': 0.8270254799165869, 'latency_sec': 0.07026720046997071}
30: {'prediction': 'yes', 'probability': 0.9669434088134082, 'latency_sec': 0.028446912765502937}
31: {'prediction': 'yes', 'probability': 0.9892910981881722, 'latency_sec': 0.024482965469360351}
32: {'prediction': 'no', 'probability': 0.3743707196074652, 'latency_sec': 0.018039226531982422}
33: {'prediction': 'yes', 'probability': 0.6942698828731126, 'latency_sec': 0.020207405090332031}
34: {'prediction': 'no', 'probability': 0.36625150556881736, 'latency_sec': 0.028900146484375}
35: {'prediction': 'yes', 'probability': 0.8771763006068757, 'latency_sec': 0.022440433502197266}
36: {'prediction': 'no', 'probability': 0.08430005537020697, 'latency_sec': 0.024910211563110351}
37: {'prediction': 'yes', 'probability': 0.6309266462026372, 'latency_sec': 0.023367643356323242}
38: {'prediction': 'yes', 'probability': 0.9647774331213098, 'latency_sec': 0.021344661712646484}
39: {'prediction': 'yes', 'probability': 0.7041087855556427, 'latency_sec': 0.022998332977294922}
40: {'prediction': 'yes', 'probability': 0.9386956803432703, 'latency_sec': 0.017781734466552734}
41: {'prediction': 'yes', 'probability': 0.7526293845138521, 'latency_sec': 0.021636962890625}
42: {'prediction': 'yes', 'probability': 0.8975508622800223, 'latency_sec': 0.023700714111328125}
43: {'prediction': 'yes', 'probability': 0.9837202873840477, 'latency_sec': 0.019345998764038086}
44: {'prediction': 'no', 'probability': 0.1794421597302704, 'latency_sec': 0.016961812973022461}
45: {'prediction': 'no', 'probability': 0.13971692248757867, 'latency_sec': 0.015748023986816406}
46: {'prediction': 'yes', 'probability': 0.945220061569987, 'latency_sec': 0.014549016952514648}
47: {'prediction': 'yes', 'probability': 0.582256922431351, 'latency_sec': 0.017320632934570312}
48: {'prediction': 'no', 'probability': 0.07798816521330144, 'latency_sec': 0.017088413238525391}
49: {'prediction': 'no', 'probability': 0.47371157675085507, 'latency_sec': 0.019417047500610351}
50: {'prediction': 'no', 'probability': 0.16689899745491213, 'latency_sec': 0.017132282257080078}
51: {'prediction': 'no', 'probability': 0.034601710133528776, 'latency_sec': 0.019313335418701172}
52: {'prediction': 'yes', 'probability': 0.8793618081039678, 'latency_sec': 0.023916482925415044}
```

```
78: {'prediction': 'no', 'probability': 0.4832047007077611, 'latency_sec': 0.020627260208129883}
79: {'prediction': 'yes', 'probability': 0.7008905003848638, 'latency_sec': 0.023018360137939453}
80: {'prediction': 'yes', 'probability': 0.9931069677636063, 'latency_sec': 0.019631147384643555}
81: {'prediction': 'yes', 'probability': 0.8151788669712864, 'latency_sec': 0.014331340789794922}
82: {'prediction': 'yes', 'probability': 0.7789080852353154, 'latency_sec': 0.015897512435913086}
83: {'prediction': 'no', 'probability': 0.43804571035149203, 'latency_sec': 0.019433259963989258}
84: {'prediction': 'yes', 'probability': 0.8946678843458649, 'latency_sec': 0.018822908401489258}
85: {'prediction': 'yes', 'probability': 0.7607789653026699, 'latency_sec': 0.016805887222290044}
86: {'prediction': 'yes', 'probability': 0.9270667039652353, 'latency_sec': 0.016531705856323242}
87: {'prediction': 'yes', 'probability': 0.9499797327755188, 'latency_sec': 0.026500940322875977}
88: {'prediction': 'yes', 'probability': 0.9889373555667279, 'latency_sec': 0.018903255462646484}
89: {'prediction': 'yes', 'probability': 0.9789721260395597, 'latency_sec': 0.017430067062377793}
90: {'prediction': 'no', 'probability': 0.29204847274986695, 'latency_sec': 0.03913402557373047}
91: {'prediction': 'yes', 'probability': 0.8433157153709434, 'latency_sec': 0.0293886661529541}
92: {'prediction': 'yes', 'probability': 0.5259115370622554, 'latency_sec': 0.034532785415649414}
93: {'prediction': 'no', 'probability': 0.1701333909946687, 'latency_sec': 0.015239715576171875}
94: {'prediction': 'no', 'probability': 0.20305650819017677, 'latency_sec': 0.018300771713256836}
95: {'prediction': 'no', 'probability': 0.11715179951792491, 'latency_sec': 0.016834020614624023}
96: {'prediction': 'yes', 'probability': 0.979441511725663, 'latency_sec': 0.01789069175720215}
97: {'prediction': 'yes', 'probability': 0.9967167886960747, 'latency_sec': 0.019572734832763672}
98: {'prediction': 'no', 'probability': 0.10662818832199962, 'latency_sec': 0.01666402816772461}
99: {'prediction': 'yes', 'probability': 0.9852346855553853, 'latency_sec': 0.02053213119506836}
100: {'prediction': 'yes', 'probability': 0.9792769516896979, 'latency_sec': 0.014632701873779297}
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
heart-api-6b57dfbc78-pl29j   1/1     Running   0          116s
heart-api-6b57dfbc78-rsnmt   1/1     Running   0          2m18s
heart-api-6b57dfbc78-sltld   1/1     Running   0          116s
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ kubectl logs heart-api-6b57dfbc78-pl29j
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
2025-12-14 12:26:08,596 | INFO | INPUT={'age': 56, 'gender': 'female', 'cp': 0, 'trestbps': 169, 'chol': 296, 'fbs': 1, 'restecg': 2, 'thalach': 123, 'exang': 1, 'oldpeak': 0.62, 'slope': 1, 'ca': 2, 'thal': 1} | PRED=no | PROB=0.3423 | LATENCY=0.2270s
INFO:     34.61.10.32:47116 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:08,646 | INFO | INPUT={'age': 62, 'gender': 'female', 'cp': 0, 'trestbps': 99, 'chol': 290, 'fbs': 0, 'restecg': 2, 'thalach': 158, 'exang': 1, 'oldpeak': 1.98, 'slope': 1, 'ca': 3, 'thal': 1} | PRED=no | PROB=0.4388 | LATENCY=0.0158s
INFO:     34.61.10.32:47120 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:08,690 | INFO | INPUT={'age': 64, 'gender': 'male', 'cp': 1, 'trestbps': 168, 'chol': 254, 'fbs': 0, 'restecg': 2, 'thalach': 106, 'exang': 1, 'oldpeak': 2.69, 'slope': 0, 'ca': 3, 'thal': 0} | PRED=no | PROB=0.1194 | LATENCY=0.0349s
INFO:     34.61.10.32:47124 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,053 | INFO | INPUT={'age': 56, 'gender': 'female', 'cp': 3, 'trestbps': 122, 'chol': 286, 'fbs': 0, 'restecg': 2, 'thalach': 153, 'exang': 0, 'oldpeak': 3.78, 'slope': 0, 'ca': 0, 'thal': 2} | PRED=yes | PROB=0.9134 | LATENCY=0.0170s
INFO:     34.61.10.32:47154 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,081 | INFO | INPUT={'age': 34, 'gender': 'male', 'cp': 1, 'trestbps': 161, 'chol': 299, 'fbs': 1, 'restecg': 0, 'thalach': 116, 'exang': 0, 'oldpeak': 3.37, 'slope': 1, 'ca': 0, 'thal': 2} | PRED=no | PROB=0.0878 | LATENCY=0.0194s
INFO:     34.61.10.32:47166 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,160 | INFO | INPUT={'age': 32, 'gender': 'male', 'cp': 3, 'trestbps': 150, 'chol': 281, 'fbs': 1, 'restecg': 0, 'thalach': 189, 'exang': 0, 'oldpeak': 2.56, 'slope': 2, 'ca': 0, 'thal': 1} | PRED=yes | PROB=0.9615 | LATENCY=0.0183s
INFO:     34.61.10.32:47192 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,183 | INFO | INPUT={'age': 74, 'gender': 'male', 'cp': 1, 'trestbps': 134, 'chol': 190, 'fbs': 1, 'restecg': 1, 'thalach': 106, 'exang': 0, 'oldpeak': 3.14, 'slope': 1, 'ca': 0, 'thal': 0} | PRED=yes | PROB=0.7832 | LATENCY=0.0158s
INFO:     34.61.10.32:47202 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,354 | INFO | INPUT={'age': 52, 'gender': 'female', 'cp': 3, 'trestbps': 100, 'chol': 206, 'fbs': 1, 'restecg': 2, 'thalach': 125, 'exang': 1, 'oldpeak': 1.74, 'slope': 2, 'ca': 3, 'thal': 0} | PRED=yes | PROB=0.9849 | LATENCY=0.0154s
INFO:     34.61.10.32:54428 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:09,743 | INFO | INPUT={'age': 54, 'gender': 'female', 'cp': 1, 'trestbps': 116, 'chol': 203, 'fbs': 1, 'restecg': 1, 'thalach': 188, 'exang': 1, 'oldpeak': 2.51, 'slope': 0, 'ca': 1, 'thal': 2} | PRED=yes | PROB=0.6330 | LATENCY=0.0524s
INFO:     34.61.10.32:54504 - "POST /predict HTTP/1.1" 200 OK
INFO:     34.61.10.32:54510 - "POST /predict HTTP/1.1" 200 OK
```

2025-12-14 12:26:10,895 | INFO | INPUT={'age': 44, 'gender': 'female', 'cp': 1, 'trestbps': 172, 'chol': 186, 'fbs': 0, 'restecg': 1, 'thalach': 181, 'exang': 0, 'oldpeak': 2.23, 'slope': 2, 'ca': 0, 'thal': 1} | PRED=yes | PROB=0.9810 | LATENCY=0.0263s
INFO: 34.61.10.32:54756 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:10,948 | INFO | INPUT={'age': 70, 'gender': 'female', 'cp': 3, 'trestbps': 149, 'chol': 163, 'fbs': 0, 'restecg': 1, 'thalach': 112, 'exang': 0, 'oldpeak': 2.92, 'slope': 1, 'ca': 3, 'thal': 1} | PRED=yes | PROB=0.9093 | LATENCY=0.0167s
INFO: 34.61.10.32:54760 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,005 | INFO | INPUT={'age': 36, 'gender': 'female', 'cp': 1, 'trestbps': 123, 'chol': 348, 'fbs': 1, 'restecg': 2, 'thalach': 105, 'exang': 1, 'oldpeak': 3.36, 'slope': 1, 'ca': 0, 'thal': 0} | PRED=yes | PROB=0.6702 | LATENCY=0.0184s
INFO: 34.61.10.32:54776 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,036 | INFO | INPUT={'age': 72, 'gender': 'female', 'cp': 2, 'trestbps': 123, 'chol': 179, 'fbs': 1, 'restecg': 0, 'thalach': 103, 'exang': 0, 'oldpeak': 3.83, 'slope': 0, 'ca': 3, 'thal': 1} | PRED=no | PROB=0.4399 | LATENCY=0.0228s
INFO: 34.61.10.32:54784 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,132 | INFO | INPUT={'age': 72, 'gender': 'male', 'cp': 0, 'trestbps': 118, 'chol': 261, 'fbs': 1, 'restecg': 2, 'thalach': 180, 'exang': 0, 'oldpeak': 0.92, 'slope': 1, 'ca': 1, 'thal': 1} | PRED=yes | PROB=0.7917 | LATENCY=0.0136s
INFO: 34.61.10.32:54814 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,181 | INFO | INPUT={'age': 33, 'gender': 'female', 'cp': 0, 'trestbps': 136, 'chol': 205, 'fbs': 0, 'restecg': 1, 'thalach': 177, 'exang': 1, 'oldpeak': 2.25, 'slope': 0, 'ca': 1, 'thal': 1} | PRED=yes | PROB=0.6333 | LATENCY=0.0164s
INFO: 34.61.10.32:54832 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,272 | INFO | INPUT={'age': 73, 'gender': 'male', 'cp': 3, 'trestbps': 138, 'chol': 262, 'fbs': 0, 'restecg': 1, 'thalach': 108, 'exang': 0, 'oldpeak': 3.75, 'slope': 2, 'ca': 0, 'thal': 0} | PRED=yes | PROB=0.9450 | LATENCY=0.0181s
INFO: 34.61.10.32:54870 - "POST /predict HTTP/1.1" 200 OK
INFO: 34.61.10.32:54894 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,333 | INFO | INPUT={'age': 56, 'gender': 'female', 'cp': 2, 'trestbps': 102, 'chol': 170, 'fbs': 0, 'restecg': 2, 'thalach': 134, 'exang': 1, 'oldpeak': 1.17, 'slope': 2, 'ca': 1, 'thal': 2} | PRED=yes | PROB=0.9597 | LATENCY=0.0199s
INFO: 34.61.10.32:54898 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,384 | INFO | INPUT={'age': 52, 'gender': 'female', 'cp': 0, 'trestbps': 104, 'chol': 292, 'fbs': 1, 'restecg': 1, 'thalach': 156, 'exang': 1, 'oldpeak': 0.33, 'slope': 0, 'ca': 1, 'thal': 0} | PRED=yes | PROB=0.8181 | LATENCY=0.0162s
2025-12-14 12:26:11,410 | INFO | INPUT={'age': 50, 'gender': 'male', 'cp': 0, 'trestbps': 167, 'chol': 259, 'fbs': 1, 'restecg': 0, 'thalach': 172, 'exang': 0, 'oldpeak': 1.31, 'slope': 0, 'ca': 1, 'thal': 2} | PRED=no | PROB=0.1278 | LATENCY=0.0174s
INFO: 34.61.10.32:54908 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,531 | INFO | INPUT={'age': 49, 'gender': 'male', 'cp': 2, 'trestbps': 146, 'chol': 243, 'fbs': 1, 'restecg': 0, 'thalach': 193, 'exang': 1, 'oldpeak': 3.21, 'slope': 1, 'ca': 0, 'thal': 1} | PRED=yes | PROB=0.7009 | LATENCY=0.0230s
INFO: 34.61.10.32:54936 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,559 | INFO | INPUT={'age': 75, 'gender': 'female', 'cp': 1, 'trestbps': 105, 'chol': 150, 'fbs': 1, 'restecg': 2, 'thalach': 122, 'exang': 1, 'oldpeak': 0.43, 'slope': 2, 'ca': 0, 'thal': 0} | PRED=yes | PROB=0.9931 | LATENCY=0.0196s
INFO: 34.61.10.32:54948 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,589 | INFO | INPUT={'age': 35, 'gender': 'female', 'cp': 0, 'trestbps': 144, 'chol': 160, 'fbs': 1, 'restecg': 2, 'thalach': 124, 'exang': 0, 'oldpeak': 0.44, 'slope': 0, 'ca': 2, 'thal': 1} | PRED=yes | PROB=0.8152 | LATENCY=0.0143s
INFO: 34.61.10.32:54954 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,719 | INFO | INPUT={'age': 35, 'gender': 'female', 'cp': 2, 'trestbps': 165, 'chol': 218, 'fbs': 1, 'restecg': 1, 'thalach': 166, 'exang': 1, 'oldpeak': 2.48, 'slope': 1, 'ca': 0, 'thal': 1} | PRED=yes | PROB=0.9271 | LATENCY=0.0165s
INFO: 34.61.10.32:54990 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,760 | INFO | INPUT={'age': 56, 'gender': 'female', 'cp': 2, 'trestbps': 104, 'chol': 188, 'fbs': 0, 'restecg': 0, 'thalach': 120, 'exang': 1, 'oldpeak': 3.12, 'slope': 2, 'ca': 1, 'thal': 0} | PRED=yes | PROB=0.9500 | LATENCY=0.0265s
INFO: 34.61.10.32:55000 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,797 | INFO | INPUT={'age': 66, 'gender': 'female', 'cp': 3, 'trestbps': 93, 'chol': 287, 'fbs': 0, 'restecg': 1, 'thalach': 162, 'exang': 1, 'oldpeak': 0.66, 'slope': 0, 'ca': 1, 'thal': 0} | PRED=yes | PROB=0.9889 | LATENCY=0.0189s
INFO: 34.61.10.32:55012 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,840 | INFO | INPUT={'age': 74, 'gender': 'female', 'cp': 3, 'trestbps': 100, 'chol': 294, 'fbs': 0, 'restecg': 1, 'thalach': 160, 'exang': 1, 'oldpeak': 1.55, 'slope': 2, 'ca': 3, 'thal': 0} | PRED=yes | PROB=0.9790 | LATENCY=0.0174s
INFO: 34.61.10.32:55016 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,889 | INFO | INPUT={'age': 55, 'gender': 'female', 'cp': 0, 'trestbps': 127, 'chol': 202, 'fbs': 1, 'restecg': 0, 'thalach': 123, 'exang': 0, 'oldpeak': 0.83, 'slope': 1, 'ca': 3, 'thal': 2} | PRED=no | PROB=0.2920 | LATENCY=0.0391s
INFO: 34.61.10.32:55026 - "POST /predict HTTP/1.1" 200 OK
2025-12-14 12:26:11,997 | INFO | INPUT={'age': 71, 'gender': 'male', 'cp': 0, 'trestbps': 154, 'chol': 225, 'fbs': 0, 'restecg': 1, 'thalach': 166, 'exang': 1, 'oldpeak': 0.28, 'slope': 0, 'ca': 3, 'thal': 1} | PRED=no | PROB=0.1701 | LATENCY=0.0152s

Deliverable 6: Performance Testing with wrk (1 MARK)

- Load testing using `wrk`
- 2000 concurrent requests
- 60-second duration
- Observed timeouts under stress
- Autoscaling triggered

Under high concurrency, latency and timeouts increase as expected. Kubernetes autoscaling helped handle the load.

```
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ wrk -t12 -c2000 -d60s -s post.lua htt
p://136.119.155.126/predict
Running 1m test @ http://136.119.155.126/predict
  12 threads and 2000 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency     0.00us    0.00us   0.00us    -nan%
    Req/Sec    10.88     13.84    90.00     86.96%
  1020 requests in 1.00m, 210.48KB read
  Socket errors: connect 983, read 0, write 0, timeout 1020
Requests/sec:     16.97
Transfer/sec:      3.50KB
```

```
[detached from 26427.wrk-test]
(base) satishsharma1911_gmail_com@instance-20251213-191643:~/Mlops_oppe2$ kubectl get pods -w
NAME                         READY   STATUS    RESTARTS   AGE
heart-api-6b57dfbc78-8ftwg   1/1     Running   0          5m21s
heart-api-6b57dfbc78-n2qj4   1/1     Running   0          5m21s
heart-api-6b57dfbc78-pl29j   1/1     Running   0          14m
```

Deliverable 7: Input Drift Detection (1 MARK)

- Training data vs inference data compared
- KS-test applied feature-wise
- Multiple features showed significant drift
- Expected due to synthetic inference data

Input drift was detected using KS-test, highlighting the importance of monitoring data distribution in production

APPLY KS-TEST FEATURE WISE

[61]:
```
numerical_features = train_df.select_dtypes(exclude="object").columns

drift_results = []

for col in numerical_features:
    train_col = train_df[col].dropna()
    inf_col = inf_df[col].dropna()

    ks_stat, p_value = ks_2samp(train_col, inf_col)

    drift_results.append({
        "feature": col,
        "ks_statistic": ks_stat,
        "p_value": p_value,
        "drift_detected": p_value < 0.05
    })

drift_df = pd.DataFrame(drift_results)
drift_df
```

[61]:

	feature	ks_statistic	p_value	drift_detected
0	age	0.247294	1.532717e-04	True
1	cp	0.241947	2.296252e-04	True
2	trestbps	0.222609	8.912579e-04	True
3	chol	0.177483	1.494938e-02	True
4	fbs	0.321485	2.052857e-07	True
5	restecg	0.336799	4.356965e-08	True
6	thalach	0.173087	1.947120e-02	True
7	exang	0.243267	2.103582e-04	True
8	oldpeak	0.407954	8.868102e-12	True
9	slope	0.200693	3.863373e-03	True
10	ca	0.392079	7.376517e-11	True
11	thal	0.683993	2.021483e-34	True

Conclusion

- End-to-end MLOps pipeline implemented
- Covers:
    - Explainability
    - Fairness
    - Deployment
    - Scaling
    - Monitoring
    - Drift detection